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Tamil Nadu Innovation Modernization through Problem Solving and Advanced Creative Tech

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DASSAULT SYSTEMES

TN-IMPACT 2026

IDENTIFYING INDUSTRIAL CHALLENGES

PROBLEM STATEMENT NUMBER : TNI26154

PROBLEM STATEMENT TITLE : AI for Safety Gear Detection-
Build a computer vision model that uses site CCTV feeds to detect workers not wearing helmets or harnesses in real-time and automatically logs the safety violation.

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Team details

Team Name : DATA MINDS

College Mentor Name : Dr. Anindita Kundu

College Name: Vellore Institute of Technology, Vellore

S.NO	STUDENT NAME	SEMESTER	DEPARTMENT
1	Ratul Pal	4	Computer Science and Engineering- Ai/Ml
2	Tarun Sagar Sah	4	Computer Science and Engineering- Blockchain
3	Dibbankur Dey	4	Computer Science and Engineering-Data Science

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Problem statement

- Construction and industrial sites are high-risk environments where strict use of safety gear (helmets, harnesses, PPE) is critical.

- Many violations go unnoticed because manual CCTV monitoring is inconsistent, time-consuming, and prone to human error.

- Current safety monitoring methods lack real-time alerts and automated violation logging for compliance tracking.

- Delayed detection of safety violations can lead to serious injuries, fatalities, legal liabilities, and financial losses.

- There is a clear need for an intelligent, automated system that continuously monitors CCTV feeds, detects missing safety equipment instantly, and logs violations for proactive safety enforcement.

Goal: Improve workplace safety through real-time AI surveillance.

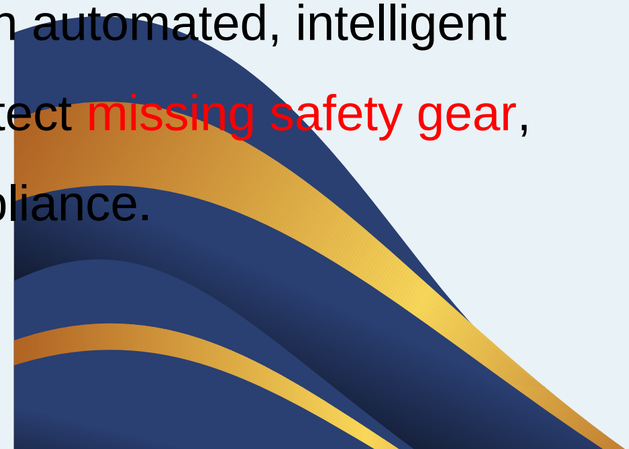
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Problem description

Workplace safety compliance on construction and industrial sites remains a major challenge despite strict regulations requiring protective gear like helmets and harnesses. In practice, continuous monitoring of workers through CCTV feeds is difficult because manual surveillance is time-consuming, inconsistent, and prone to human error. As a result, safety violations often go unnoticed. This lack of real-time detection increases the risk of accidents, injuries, and legal liabilities. Therefore, there is a strong need for an automated, intelligent system that can continuously monitor live video feeds, instantly detect missing safety gear, and reliably log violations to improve safety enforcement and compliance.



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Objectives

- To develop a computer vision model that analyzes live CCTV footage from construction sites in real time.
- To accurately identify instances where required safety gear, such as helmets or harnesses, is missing.
- To achieve high detection accuracy suitable for real-world site deployment.
- To automatically log detected safety violations with relevant data for monitoring and compliance.
- To trigger immediate alerts so that corrective action can be taken without delay.



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PROPOSED SOLUTION — SYSTEM PIPELINE

- **Camera Input:** Multiple camera feeds capture live site footage. Frames are sampled periodically for efficient processing.
- **AI Detection:** Each frame is analyzed using a trained PPE detection model to identify persons, helmets, and safety vests.
- **Risk Tracking:** A risk engine monitors detected violations over time and calculates risk scores for each worker.
- **Alerts & Logging:** When violations persist, alerts are generated and recorded along with timestamps and evidence frames.

In one line:

Cameras → AI Detection → Risk Analysis → Automated Alerts



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System Overview- Main Modules

DATA ACQUISITION

📷 Camera Capture Module
Captures multiple camera streams and samples frames once per second for efficient processing..

React Frontend
Displays live feeds, supports multi-camera view, sends frames to backend, overlays detections, and handles violation evidence storage and alerts.

INTELLIGENCE CORE

FastAPI Backend
Receives frames, decodes images, runs detection pipeline, manages logic, updates violation records, and returns JSON predictions.

🦺PPEDetector Model
YOLOv8-based model that detects persons, helmets, and vests with bounding boxes and confidence scores.

📊Risk Engine
Tracks violations per person using timestamps, maintains event history, and computes risk scores over time.

Deployment & Output

🚨Violation Alert System
Stores first violation frame, triggers notifications, and maintains overlays until violations are resolved.

📦Packaged Executable Module
Combines frontend, backend, model, and utilities into a single plug-and-play executable.

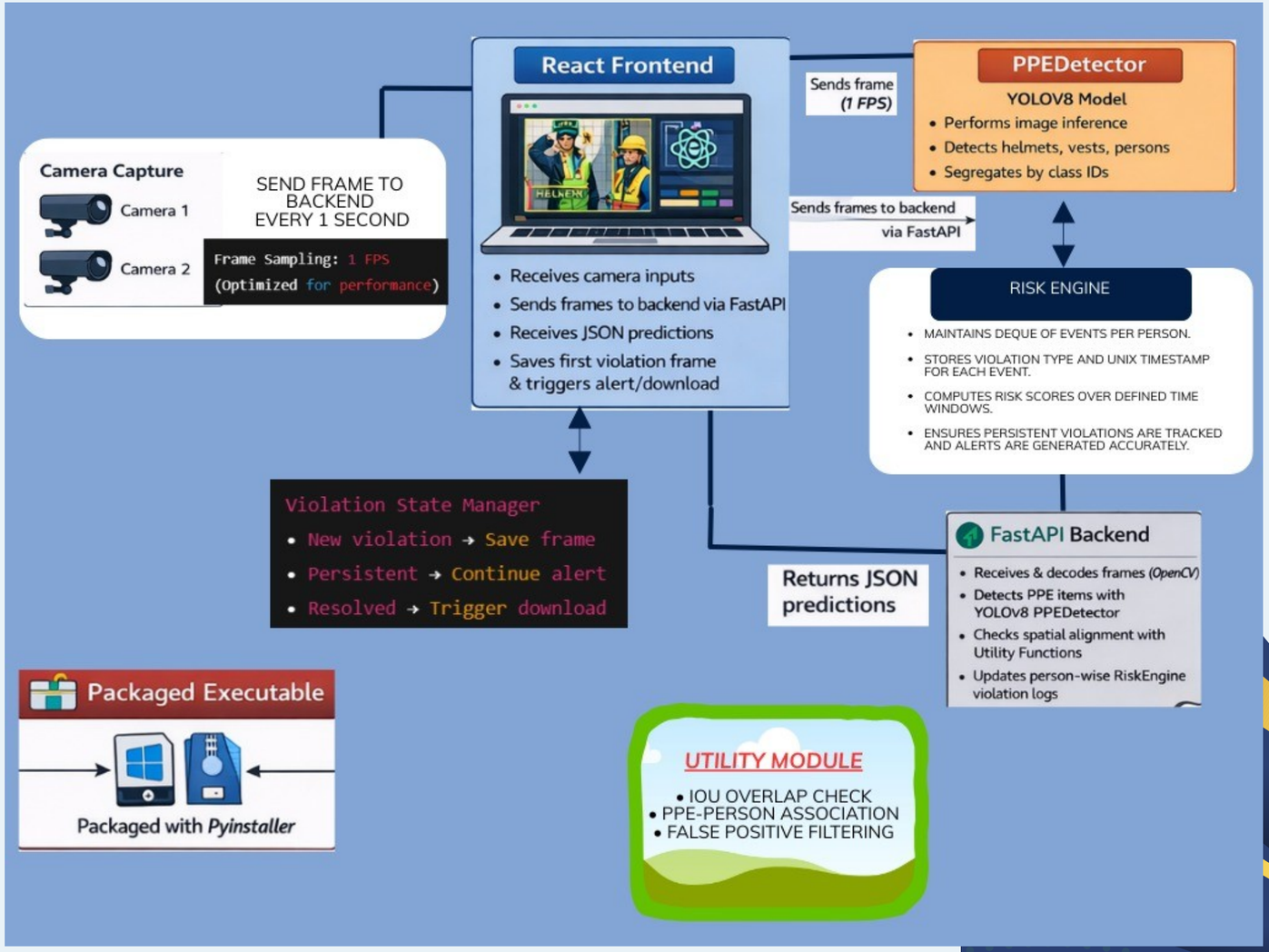
EACH MODULE WORKS TOGETHER TO DELIVER REAL-TIME, AUTOMATED SAFETY MONITORING.

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ARCHITECTURE DIAGRAM



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WORKFLOW PIPELINE — SYSTEM FLOW

CAPTURE → **SEND** → **DETECT** → **EVALUATE** → **RETURN** → **ALERT**

CAPTURE	SEND	DETECT	EVALUATE	RETURN	ALERT
Frames sampled from multiple camera streams.	Frontend sends frames to backend via HTTP POST.	YOLOv8 model detects persons and PPE items.	Risk Engine analyzes violations over time.	Backend sends JSON results to frontend.	Frontend shows overlays and triggers violation notifications.



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ML-MODEL

Model: YOLOv8 (Ultralytics)

Classes Detected: Person, Helmet, No-Helmet, Vest (in place of harness) (IDs: 0, 2, 6, 7)

Confidence Threshold: 0.20 — set low to minimize missed detections in safety-critical context

Why YOLOv8: Real-time detection, lightweight enough for deployment, strong accuracy on PPE datasets



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Backend Logic

Frame Decoding: Receives image bytes via POST /analyze, decodes with OpenCV into BGR frame
Detection Processing: YOLO runs inference → outputs class ID, confidence, bounding box (xy xy) per detection

PPE Association:
Helmet → if helmet center lies inside person box AND is in upper half (head region)
Vest → person if bounding boxes overlap

Violation Logic:
No helmet detected near person's head OR explicit no_helmet class found → **helmet violation**
No vest overlap found **→ vest violation**

Risk Tracking: Risk Engine instances the logs violations with timestamps using a deque
Response: Returns person count + violation counts

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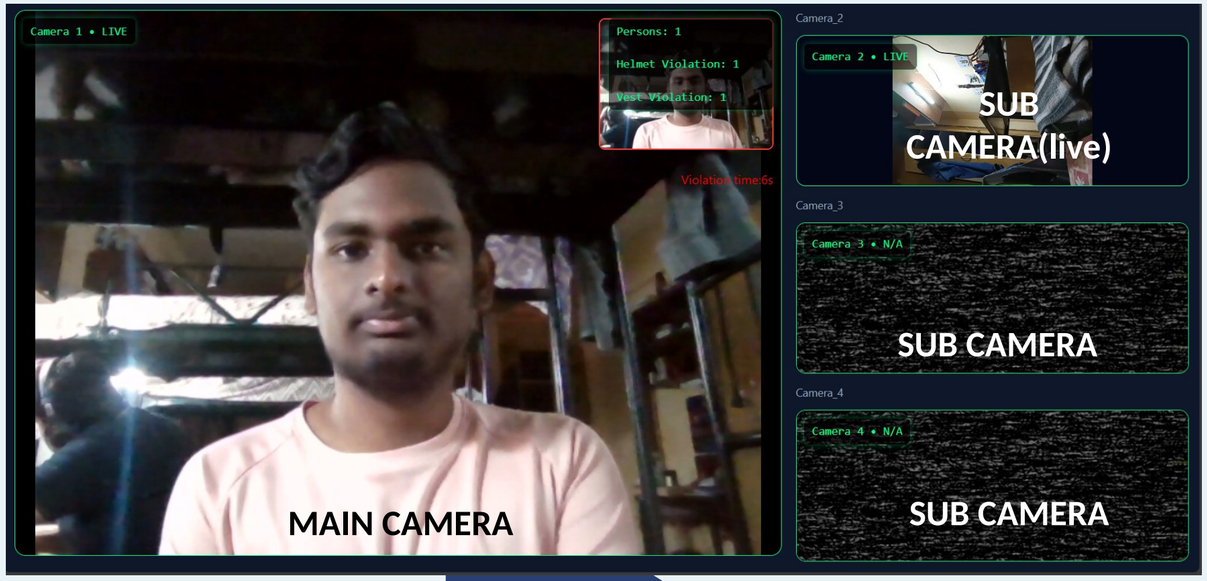
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FRONTEND-SYSTEM

FRONTEND OVERVIEW

- Sends frames to backend
- Live multi-camera dashboard
- Displays real-time detections



FRONTEND WORKING

⏱️ Frame sampling (1 FPS)

⚡ API communication

🔔 Overlay rendering

⚠️ Alert trigger system

VIOLATION HANDLING

Saves first violation frame and triggers alert/download once violation is resolved.

The frontend acts as the real-time control panel connecting cameras, AI detection, and alerts.

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OPTIMIZATION STRATEGY — PERFORMANCE DESIGN CHOICES

Frame Sampling (1 FPS)
Processing every frame is computationally heavy. Sampling once per second reduces load while still detecting violations reliably.

Efficient Model Inference
Detection runs only on sampled frames, ensuring real-time responsiveness without GPU overload.

Lightweight Data Transfer

[Full Video] →[Compressed BLOB] → [Fast Backend]

Selective Logging
Only violation frames are stored, reducing storage usage and avoiding unnecessary data accumulation.



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DEMO— Live System in Action

🎬 Recorded Demo

https://youtu.be/gSikuVNupWM

https://youtu.be/zrvYNF5XT34

⚡ LIVE EXECUTION

- Run system from USB
- Real-time detection
- Instant alerts

⬇️ Deployment Access

https://drive.google.com/drive/folders/1Mh6IURDg4hLgYRlpoS8jp2zb3SqvBxwS?usp=sharing



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Findings

- The system successfully performs **real-time PPE detection** using a YOLOv8-based model with reliable accuracy at low confidence thresholds.
- Frame sampling at **1 FPS** provides an effective balance between performance and computational efficiency, enabling smooth real-time monitoring.
- The **RiskEngine tracking mechanism** accurately identifies persistent violations by logging timestamped events per person.
- Spatial utility functions significantly **reduce false positives**, ensuring PPE is correctly assigned only to relevant workers.
- Multi-camera support demonstrates **scalability** for real construction site environments.
- Packaging the entire system into a **single executable** enables easy deployment without dependency installation,
- making it practical for on-site usage.

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Implementation Links

- Frontend Repository → Click Here
- Backend Repository → Click Here
- Deployed Frontend(Static preview) →Click Here
- Demo Video #1→ Click Here
- Demo Video #2→ Click Here

- Executable Download → Click Here
  - Instructions:
    - Extract the ZIP file
    - **Navigate to:** CONSTRUCTION-PPE-DETECTION-main→dist→run→run.exe
    - Run **run.exe** (≈ 48 MB)



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Conclusion

The developed system demonstrates a complete end-to-end solution for automated **safety compliance monitoring**. By integrating **real-time computer vision**, **intelligent violation tracking**, and a **responsive frontend interface**, the project successfully replaces manual monitoring with an **accurate, scalable, and deployable system**.

The system is packaged into a **single executable file**, allowing it to run instantly without requiring any additional installations or setup, making the solution not only technically robust but also **industry-ready**. This approach highlights the potential of AI-driven monitoring systems to **improve workplace safety**, **reduce human error**, and enable proactive **risk management**.